Zhe (Jay) Xu

Research Scientist | LLMs & Generative Modeling

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Research Profile

Research scientist working on large language models (LLMs), diffusion generative modeling, and their intersection (diffusion LLMs). Current research develops LLM post-training recipes for reasoning, ranking, and retrieval at Meta. Doctoral research focused on graph machine learning, including graph generation, dataset distillation, and many more.

Education

University of Illinois Urbana-Champaign *Urbana, IL*
Ph.D. (2025) and M.S. (2021) in Computer Science

Fudan University *Shanghai, China*
B.E. in Electronic Engineering, 2018

Work Experience

Meta, Monetization Sunnyvale, CA
Research Scientist; previously Research Intern (2024) *2025–Present*

- Develop LLM post-training recipes for ads retrieval and re-ranking.
  - Lead semantic ID development and contribute to mid-training for a generative reasoning re-ranker.
  - Develop LLM-native two-tower retrieval models.
  - Develop permutation diffusion LLMs for generative re-ranking.
- Lead academic collaborations and mentor student researchers.
  - Establish university collaborations on LLM test-time learning, diffusion LLM decoding, reinforcement learning (RL) data attribution, and LLM over-scope benchmarking, serving as the industry research mentor.
  - Serve as intern manager and research mentor for the agentic continual learning project.

Visa Research — Research Intern; graph data augmentation 2023

Meta, Applied Machine Learning — Research Intern; large-scale graph-based recommendation 2022

Selected Publications

*Equal contribution. †Research Mentor. Complete publication list available at Google Scholar.

Preprints

Zhe Xu, et al. *Exploring the Design Space of Permutation Diffusion LLMs.* Preprint.

Zhe Xu, et al. *The Case Against Generation for Retrieval: Discriminative Language Models as Effective Retrievers.* Preprint.

Mingfu Liang*, Yufei Li*, **Jay Xu***, et al. *Generative Reasoning Re-ranker.* Preprint.

Xinyu Zhu, **Zhe Xu**†, et al. *Self-Guided Test-Time Training for Long-Context LLMs.* Preprint.

Haofei Yu, **Zhe Xu**†, et al. *Rethinking Agentic Continual Learning as Bayesian Discovery.* Manuscript.

Fengwei Teng, **Zhe Xu**, et al. *Learning Parallel Decoding from Multiple Samplers in Diffusion Language Models.* Manuscript.

Roman Makarov, **Zhe Xu**, et al. *Fast at the Frontier, Wrong at the Fallback: A Commit-Level Analysis of Threshold Decoding in Diffusion Language Models.* Manuscript.

Peer-Reviewed Publications

Zhe Xu, et al. *Tokenizing Numerical and Embedding Features for LLM RecSys.* **TokShop at COLM**, 2026.

Zhe Xu, et al. *How to Make LMs Strong Node Classifiers?.* **Findings of EACL**, 2026.

Zhe Xu, et al. *Fine-grained Graph Rationalization.* **CIKM**, 2025.

Ruizhong Qiu, **Zhe Xu**, et al. *Turing Completeness of Prompting.* **ICLR**, 2025.

Dongqi Fu*, Liri Fang*, Zihao Li*, **Zhe Xu***, et al. *Bring Complex Geometric Information to LLMs: A Positional Survey of Graph Parametric Representation.* **LOG**, 2025.

Zhe Xu, et al. *Discrete-state Continuous-time Diffusion for Graph Generation.* **NeurIPS**, 2024.

Huiyuan Chen, **Zhe Xu**, et al. *Masked Graph Transformer for Large-Scale Recommendation.* **SIGIR**, 2024.

Zhe Xu, et al. *Kernel Ridge Regression-Based Graph Dataset Distillation.* **KDD**, 2023.

Zhe Xu, et al. *Node Classification Beyond Homophily: Towards a General Solution.* **KDD**, 2023.

Zhe Xu, et al. *Generalized Few-Shot Node Classification on Graphs: Toward an Uncertainty-Based Solution.* **KAIS**, 2023.

Zhe Xu, et al. *Generalized Few-Shot Node Classification on Graphs.* **ICDM (Best-Ranked Papers)**, 2022.

Zhe Xu, et al. *Graph Sanitation with Application to Node Classification.* **The Web Conference**, 2022.

Kaize Ding, **Zhe Xu**, et al. *Data Augmentation for Deep Graph Learning: A Survey.* **SIGKDD Explorations**, 2022.

Zhe Xu, et al. *DESTINE: Dense Subgraph Detection on Multi-Layered Networks.* **CIKM**, 2021.

Zhe Xu, et al. *Ranking on Network of Heterogeneous Information Networks.* **IEEE BigData**, 2020.

Dongqi Fu*, **Zhe Xu***, et al. *A View-Adversarial Framework for Multi-View Network Embedding.* **CIKM**, 2020.

Selected Talks

Invited Talk, National University of Singapore AI Institute. *Generative Reasoning Re-ranker.* 2026.

Invited Talk, Visa Research. *Graph Data Distillation.* 2023.

Lecture-style tutorial, WSDM. *Natural and Artificial Dynamics in Graph Neural Networks: A Tutorial.* 2023.